

10 (a)	electromagnetic radiation/photons incident on a surface	B1	
	causes emission of electrons (from the surface)	B1	[2]
(b)	$E = hc / \lambda$		
	$= (6.63 \times 10^{-34} \times 3.00 \times 10^8) / (436 \times 10^{-9})$	C1	
	$= 4.56 \times 10^{-19} \text{ J } (4.6 \times 10^{-19} \text{ J})$	A1	[2]
(c) (i)	$\Phi = hc / \lambda_0$		
	$\lambda_0 = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / (1.4 \times 1.60 \times 10^{-19})$	C1	
	$= 890 \text{ nm}$	A1	[2]
(ii)	$\lambda_0 = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / (4.5 \times 1.60 \times 10^{-19})$		
	$= 280 \text{ nm}$	A1	[1]
(d)	caesium:		
	wavelength of photon less than threshold wavelength (or v.v.)		
	or		
	$\lambda_0 = 890 \text{ nm} > 436 \text{ nm}$		
	so yes	A1	
	tungsten:		
	wavelength of photon greater than threshold wavelength (or v.v.)		
	or		
	$\lambda_0 = 280 \text{ nm} < 436 \text{ nm}$		
	so no	A1	[2]